

Hybrid cars are those that run on both a rechargeable battery and fuel. There are also some hybrids that use petrol and another form of fuel: biodiesel, created from the Jatropha plant that grows only in certain areas of the world—including India.

There are two types of hybrid cars—parallel and serial. The parallel hybrid has a petrol engine that runs off a fuel tank, and also has an electric engine that is powered by a set of batteries. The advantage is that the electric engine can turn the transmission and take care of initial acceleration, without having the petrol engine do any work.

The other kind of hybrid—the series hybrid—uses a petrol engine to power a generator. This generator can either charge the batteries or power the electric motor that drives the transmission. Thus, in a series hybrid, the petrol engine never directly turns the transmission.

Hybrid Components

A hybrid car contains the following parts:

▣ **Petrol engine:** A hybrid car's petrol engine is similar to regular ones, but these engines are smaller and uses advanced technologies to reduce emissions and increase efficiency.

▣ **Electric motor:** The electric motor powering a hybrid car is very sophisticated. Advanced electronics allow it to act as a motor cum generator. Thus, the motor can draw energy from the batteries to accelerate the car, and also act as a generator and return energy to the batteries when it slows down.

This also happens to be the biggest difference between a hybrid car and fully electric car, as the latter needs to be recharged often, whereas the battery in a hybrid car recharges when the petrol engine is being used, or when the electric motor acts as a generator.

▣ **Batteries:** The batteries in a hybrid car are the energy storage device for the electric motor. Unlike the petrol engine, which only draws fuel from the tank, the electric motor can draw energy from the batteries as well as put energy back. Actually, it's a set of nickel metal hydride batteries that the electric engine draws its power from.

▣ **Design:** While the car is perfectly set under the hood, what's covering it is equally important. A frame built for maximum aerodynamics and minimum requisite weight will ensure the best fuel efficiency and a massive reduction in emissions.

Computer Control

For the entire ride, a hybrid car's inbuilt computer calculates when to let the petrol engine do the work, and the kind of boost it needs from the electric motor. The onboard computer is also actively involved in calculating and deciding when to reclaim excess energy while driving using the electric motor.

The computer also monitors the amount of charge in the batteries, ensuring they are always between 30 and 80 per cent charged. This allows the batteries to last almost 3,000 to 3,500 km—not bad at all!

In the Toyota Prius, a hybrid car, when you accelerate, you really control a device that tells

Concept Hybrids

Like with any new car that's built from the ground up, a new hybrid has to be constructed at the drawing board first. Here's some information from the Net about some concept cars.

Volvo 3CC

Designers, engineers and business people at the Volvo Monitoring and Concept Center think-tank in California came up with the 3CC when confronted with the task of creating a 'future-proof concept' that would enhance sustainable mobility. The idea was to make a car that was not only fuel-efficient, versatile, comfortable and safe, but also exciting to drive and look at.

In the 3CC, Volvo aimed to attractively present a concept that



focused on efficient mobility. This was achieved through good aerodynamics on a compact footprint, light-weight body material, and an electric powertrain. Volvo opted to give the 3CC a high strength steel space frame and composite sandwich floor panels for safety and reduced weight. In fact, the outer body is a bonded one piece carbon fibre shell.

With a potential driving range of over 300 km under certain driving conditions, the torque-to-weight ratio is roughly comparable to the V70 T5 but available over 0–3,500 rpm.

This performance is achieved using a drivetrain that has been specifically designed for the 3CC, although it, too, is a prototype. The electric power comes from lithium-ion cells that are identical to those used in modern laptop computers that are packaged in the thin sandwich floor.

Toyota Volta

Way back in 1800, Count Alessandro Volta arranged zinc and copper discs in a column and invented the battery. Circa 2004, Toyota electrified automotive history with the first high-performance



hybrid, named in his honour.

The Giugiaro-designed carbon-fibre body seats three people abreast and features "drive-by-wire" controls, allowing you to position the steering wheel and pedals in front of any one of them. And the Volta's 408-hp Hybrid Synergy Drive (a 3.3-litre V6 with an electric motor for each axle) not only delivers 700 km on a 52 litre tank, but also facilitates 0-80 kmph acceleration in about four seconds.

Operating electronics enable the power transmitted by the electric motors to the wheels to be modulated, thereby rendering superfluous the gear/clutch unit. Housed under the lightweight, carbon-fibre chassis are the batteries, weighing 70 kg.